

GENERAL DESCRIPTION

The adc1395x is a CMOS 3.3V 10-bit analog-to-digital converter (ADC). It converts the analog input signal into 10-bit binary digital codes at a maximum conversion rate of 500KSPS with 2.5MHz clock.

The device is a recycling type monolithic ADC with an on-chip sample-and-hold function. The ADC has power down mode.

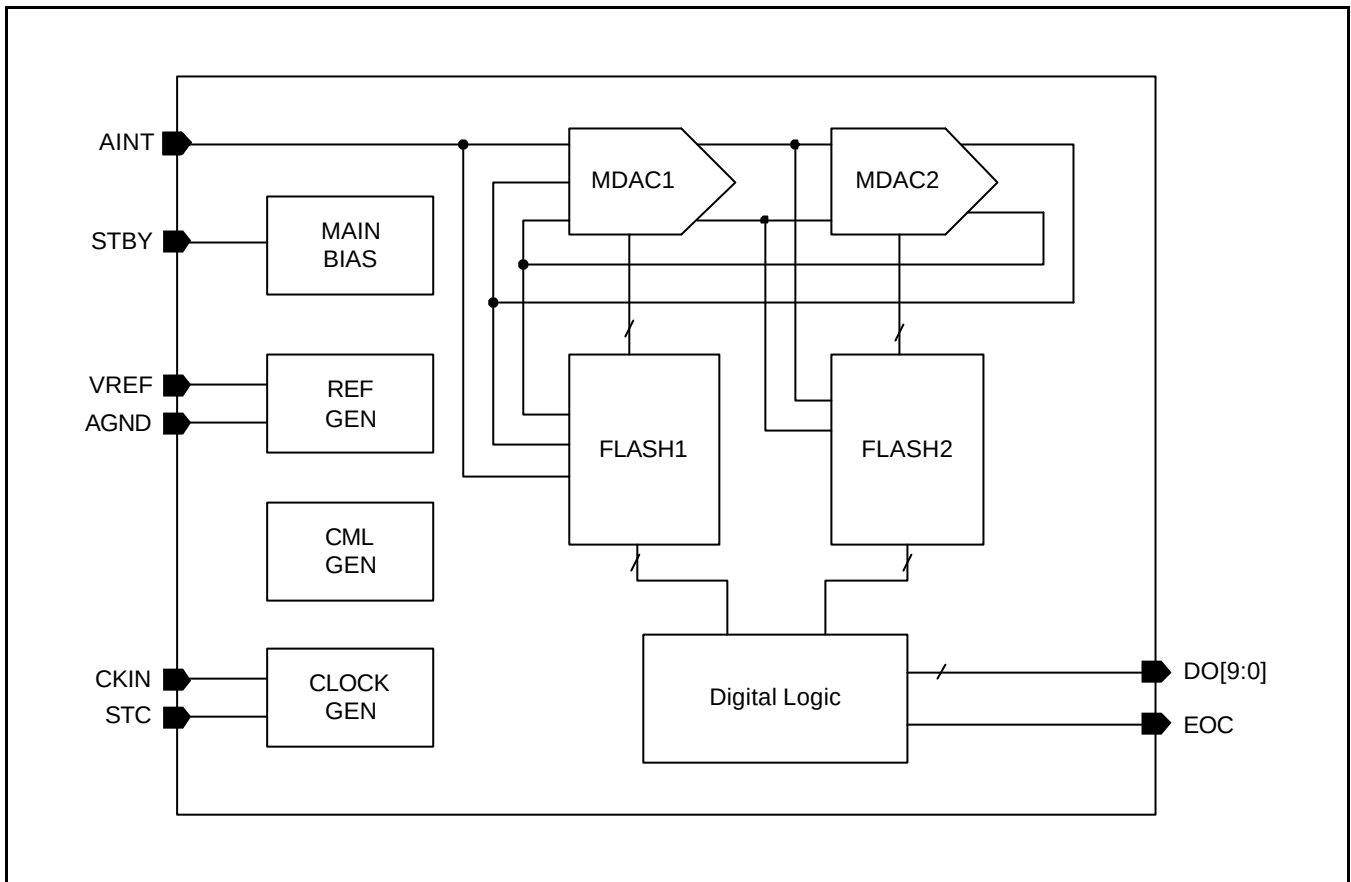
FEATURES

- Resolution: 10-bit
- Maximum conversion rate: 500KSPS
- Main clock: 2.5MHz (max.)
- Power supply: 3.3V $\pm$ 0.3V
- Total current: 20 μ A (Standby Mode)
2.0mA (Normal Operation)
- Input range: 0.0V ~ 3.3V (3.3V P-P)
- Differential linearity error: $\pm$ 1.0 LSB (Max)
- Integral linearity error: $\pm$ 2.0 LSB (Max)
- Signal to noise & distortion ratio: 54dB (Typ.)
- Digital output: CMOS Level
- Operating temperature range: -40°C ~ 85°C

TYPICAL APPLICATIONS

- MICOM interface
- Portable equipment
- Low-voltage low-power application

FUNCTIONAL BLOCK DIAGRAM



Ver 1.1 (Feb. 2004)

No responsibility is assumed by SEC for its use nor for any infringements of patents or other rights of third parties that may result from its use. The content of this data sheet is subject to change without any notice.

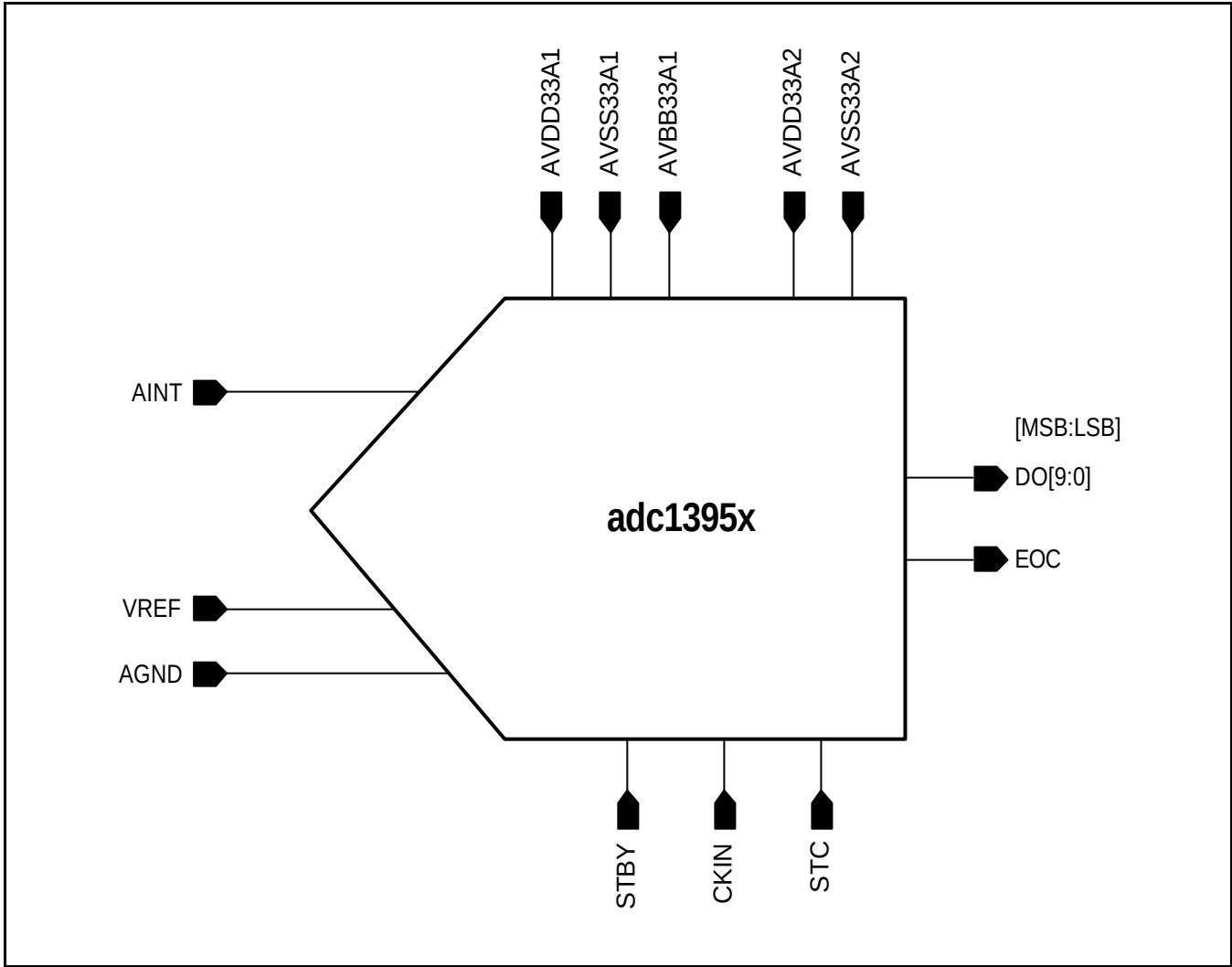
CORE PIN DESCRIPTION

Name	I/O Type	I/O Pad	Pin Description
VREF	AI	phia_abb	Reference top (3.3V)
AGND	AI	phia_abb	Reference bottom (0.0V)
AVDD33A1	AP	vdd3th_abb	Analog power (3.3V)
AVBB33A1	AG	vbbh_abb	Analog sub bias (0.0V)
AVSS33A1	AG	vssth_abb	Analog ground (0.0V)
AIN _T	AI	phiar50_abb	Analog input (input range: 0.0V ~ 3.3V)
STBY	DI	phicc_abb	VDD = Power saving (standby), GND = Normal
CKIN	DI	phicc_abb	Sampling clock input
DO[9:0]	DO	phob4_abb	Digital output
EOC	DO	phob4_abb	End of conversion signal
STC	DI	phicc_bb	Start of conversion signal
AVSS33A2	DG	vssth_abb	Digital GND (0.0V)
AVDD33A2	DP	vdd3th_abb	Digital power (3.3V)

I/O Type Abbr.

- AI: Analog Input
- DI: Digital Input
- AO: Analog Output
- DO: Analog Output
- AP: Analog Power
- AG: Analog Ground
- DP: Digital Power
- DG: Digital Ground
- AB: Analog Bi-Direction
- DB: Digital Bi-Direction

CORE CONFIGURATION



ABSOLUTE MAXIMUM RATINGS

Characteristics	Symbol	Value	Unit
Supply voltage	VDD	4.8	V
Analog input voltage	AINT	VSS to VDD	V
Digital input voltage	CKIN	VSS to VDD	V
Reference voltage	VREF / AGND	VSS to VDD	V
Storage temperature range	Tstg	-45 to 150	°C
Operating temperature range	Topr	-40 to 85	°C

NOTES:

1. Absolute maximum rating specifies the values beyond which the device may be damaged permanently. Exposure to absolute maximum rating conditions for extended periods may affect reliability. Each condition value is applied with the other values kept within the following operating conditions and function operation under any of these conditions is not implied.
2. All voltages are measured with respect to VSS unless otherwise specified.
3. 100pF capacitor is discharged through a 1.5k Ω resistor (Human body model)

RECOMMENDED OPERATING CONDITIONS

Characteristics	Symbol	Min	Typ	Max	Unit
Supply voltage	AVDD33A1 AVDD33A2	3.0	3.3	3.6	V
Reference input voltage	VREF AGND	2.0 0.0	3.3 0.0	3.6 0.0	V
Analog input voltage	AINT	0.0	VREF	-	V
Operating temperature	Toper	-40	-	85	°C

NOTE: It is strongly recommended that all the supply pins (AVDD33A1, AVDD33A2) be powered from the same source to avoid power latch-up.

DC ELECTRICAL CHARACTERISTICS

Characteristics	Symbol	Min	Typ	Max	Unit	Test Condition
Differential nonlinearity	DNL	-	± 0.8	± 1	LSB	VREF = 3.3V AGND = 0.0V
Integral nonlinearity	INL	-	± 1	± 2	LSB	VREF = 3.3V AGND = 0.0V
Offset voltage	OFF	-	3	8	LSB	VREF = 3.3V AGND = 0.0V

(Converter Specifications: AVDD33A1 = AVDD33A2 = 3.3V, AVSS33A1 = AVSS33A2 = 0V, Toper = 25°C, VREF = 3.3V, AGND = 0.0V unless otherwise specified)

AC ELECTRICAL CHARACTERISTICS

Characteristics	Symbol	Min	Typ	Max	Unit	Test Condition
Maximum conversion rate	fc	-	-	500	KSPS	f _{CKIN} = 2.5MHz
Standby supply current	-	-	20	60	uA	STBY = VDD
Dynamic supply current	IVDD	-	2.0	3	mA	f _{CKIN} = 2.5MHz (without system load)
Reference current	IREF	-	0.4	0.6	mA	V _{REF} = 3.3V
Total harmonic distortion	THD	-	-60	-56	dB	f _{CKIN} = 2.5MHz AINT = 100kHz
Signal-to-noise & distortion ratio	SNDR	50	54	-	dB	f _{CKIN} = 2.5MHz AINT = 100kHz

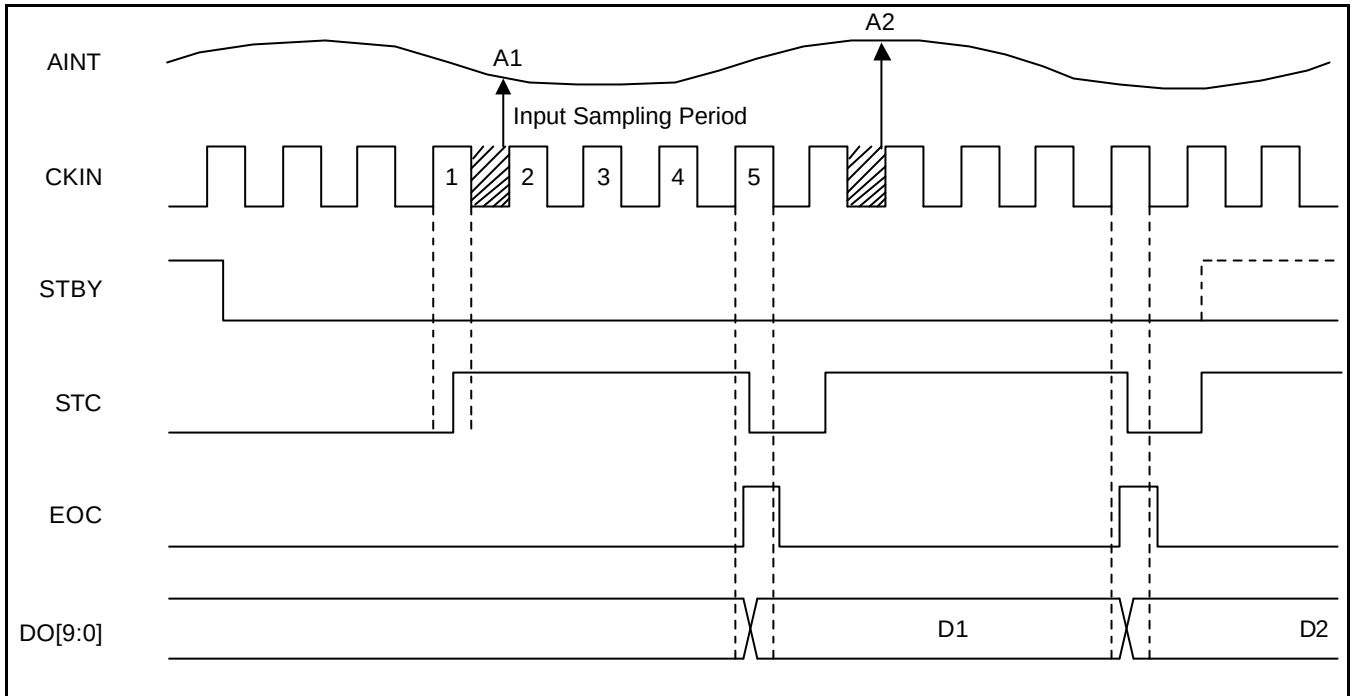
(Converter Specifications: AVDD33A1 = AVDD33A2 = 3.3V, AVSS33A1 = AVSS33A2 = 0V, Toper = 25°C, VREF = 3.3V, AGND = 0.0V unless otherwise specified)

I/O CHART

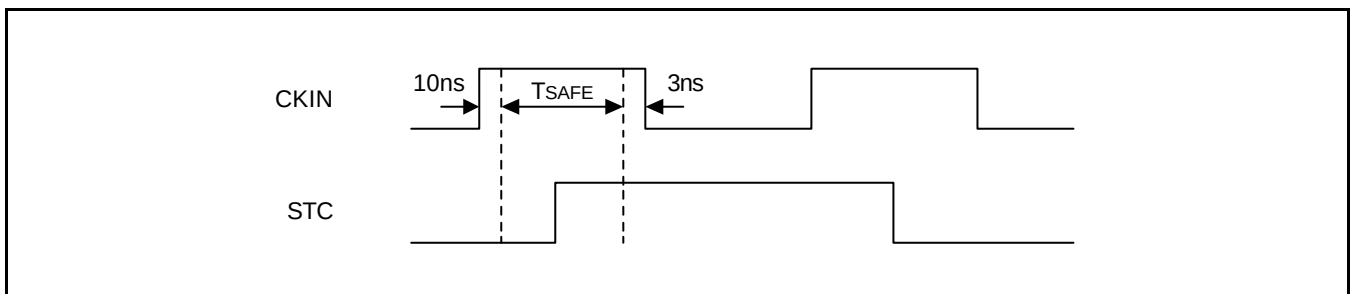
Index	AINT Input (V)	Digital Output	
0	~ 0.00322	00 0000 0000	1LSB = 3.22mV VREF = 3.3V AGND = 0.0V
1	0.00322 ~ 0.00644	00 0000 0001	
2	0.00644 ~ 0.00967	00 0000 0010	
~	~	~	
511	1.64678 ~ 1.65000	01 1111 1111	
512	1.65000 ~ 1.65322	10 0000 0000	
513	1.65322 ~ 1.65644	10 0000 0001	
~	~	~	
1021	3.29033 ~ 3.29355	11 1111 1101	
1022	3.29355 ~ 3.29678	11 1111 1110	
1023	3.29678 ~	11 1111 1111	

TIMING DIAGRAM

1. Main Waveform



2. STC & CKIN Condition



The A/D Converter operates data conversion when STC (Start Conversion) signal is just "HIGH". Otherwise, output data (DO[9:0]) keep the current states. The STC signal should be changed during "T_{SAFE}" with the "HIGH" level of the clock to operation as shown in the main waveform.

· ADC External Interface Signal

AINT: Analog Input Signal (Input)

Input Range: VREF ~ AGND

STBY: Stand-by Signal, Power Save Mode (Input)

CKIN: ADC Main Clock, $f_{CKIN} = 2.5\text{MHz}$, 1 Clock Period = 400ns (Input)

STC: Start of Conversion Signal (Input)

EOC: End of Conversion Signal (Output)

DO[9:0]: Digital Output Signal (Output)

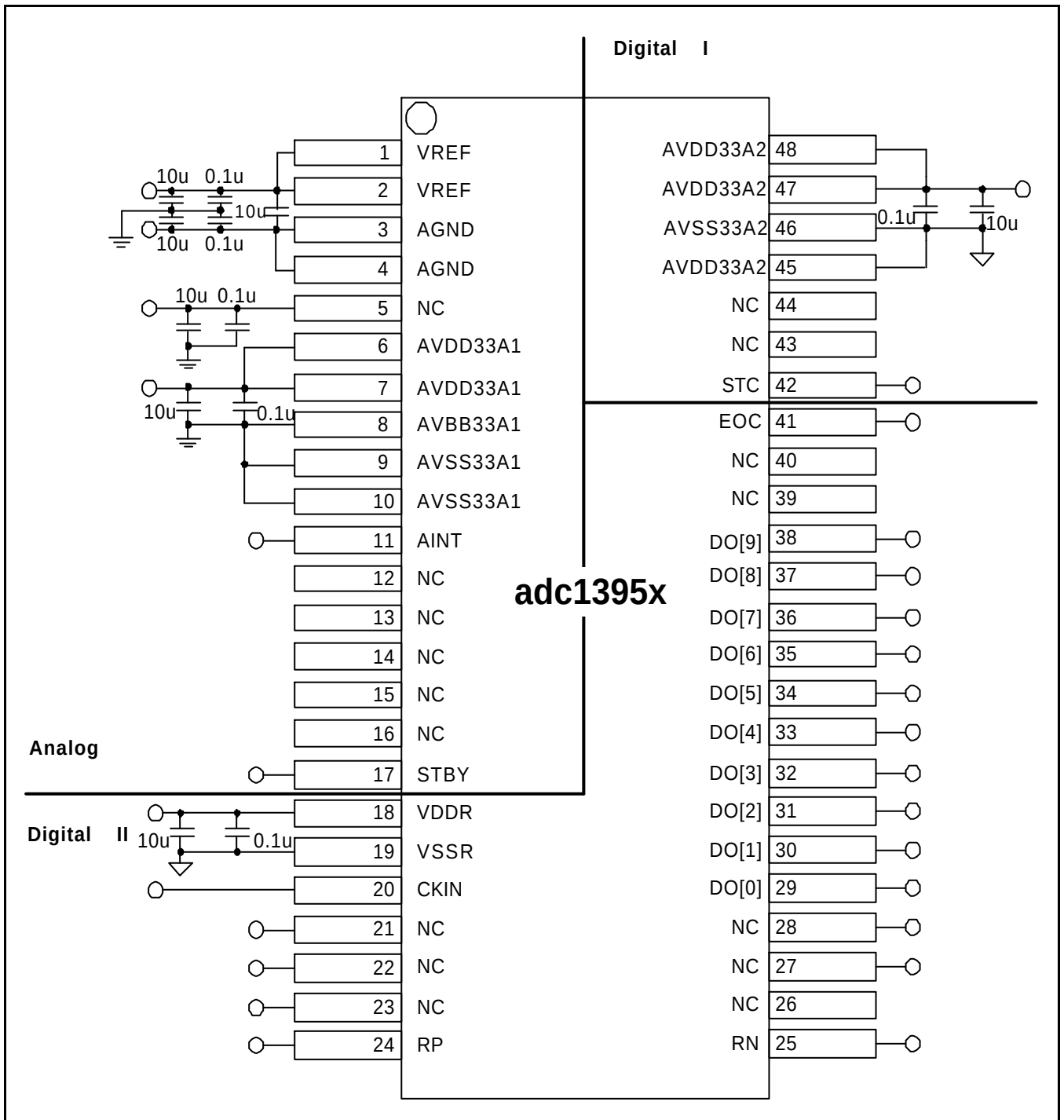
ADC function is evaluated by external check on the bi-directional pads connected to input nodes of HOST DSP back-end circuit.

The diagram illustrates the ADC1395x block and its connections. The central component is the **adc1395x**, which has several inputs and outputs:

- Inputs:**
 - AIN_T**: Analog input terminal.
 - VREF**: Reference voltage input.
 - AGND**: Analog ground input.
 - AVDD33A1**, **AVSS33A1**, **AVBB33A1**: 3.3V analog power and ground pins.
 - AVDD33A2**, **AVSS33A2**: 3.3V analog power and ground pins.
 - STBY**: Standby control pin.
 - CKIN**: Clock input pin.
 - STC**: Start of Conversion pin.
- Outputs:**
 - DO[9:0]**: 10-bit digital output data bus, labeled **[MSB:LSB]**.
 - EOC**: End of Conversion signal.

The digital output **DO[9:0]** is connected to a **Digital Mux**, which in turn connects to the **HOST DSP CORE**. A **Bi-directional PAD** is also connected to the **Digital Mux** and the **HOST DSP CORE**, with a bidirectional data path labeled **D[9:0]**. A note at the bottom indicates: **(ADC Function Test & Externally Forced Digital Input)**.

PACKAGE CONFIGURATION

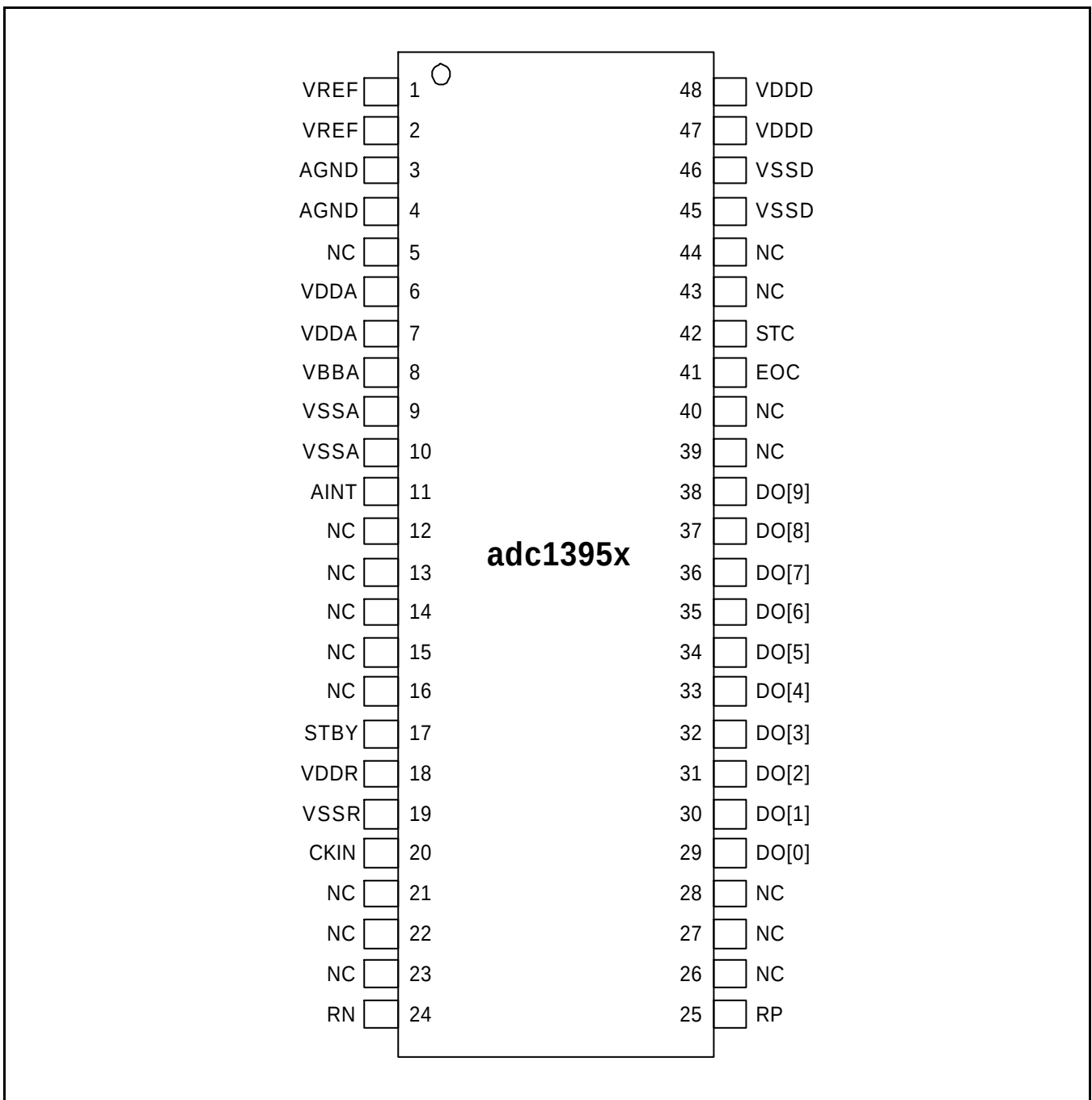


NOTE: NC denotes "No Connection".

PACKAGE PIN DESCRIPTION

Name	No.	I/O Type	Pin Description
VREF	1, 2	AI	Reference voltage (3.3V)
AGND	3, 4	AI	Analog ground (0.0V)
AVDD33A1	6, 7	AP	Analog power (3.3V)
AVBB33A1	8	AG	Analog sub bias
AVSS33A1	9, 10	AG	Analog ground
AIN _T	11	AI	Analog input
STBY	17	DI	VDD = Power saving (Standby), GND = Normal
VDDR	18	PP	PAD power (3.3V)
VSSR	19	PG	PAD ground
CKIN	20	DI	Clock input ($f_{CKIN} = 2.5\text{MHz}$)
RP	24	AO	Test pin1
RN	25	AO	Test pin2
DO[0]	29	DO	Digital output (LSB)
DO[1:8]	30 ~ 37	DO	Digital output
DO[9]	38	DO	Digital output (MSB)
EOC	41	DO	End of conversion signal
STC	42	DI	Start of conversion signal
AVSS33A2	45, 46	DG	Digital GND
AVSS33A2	47, 48	DP	Digital power (3.3V)

PACKAGE CONFIGURATION



NOTE: I/O TYPE PP and PG denote PAD Power and PAD Ground respectively.

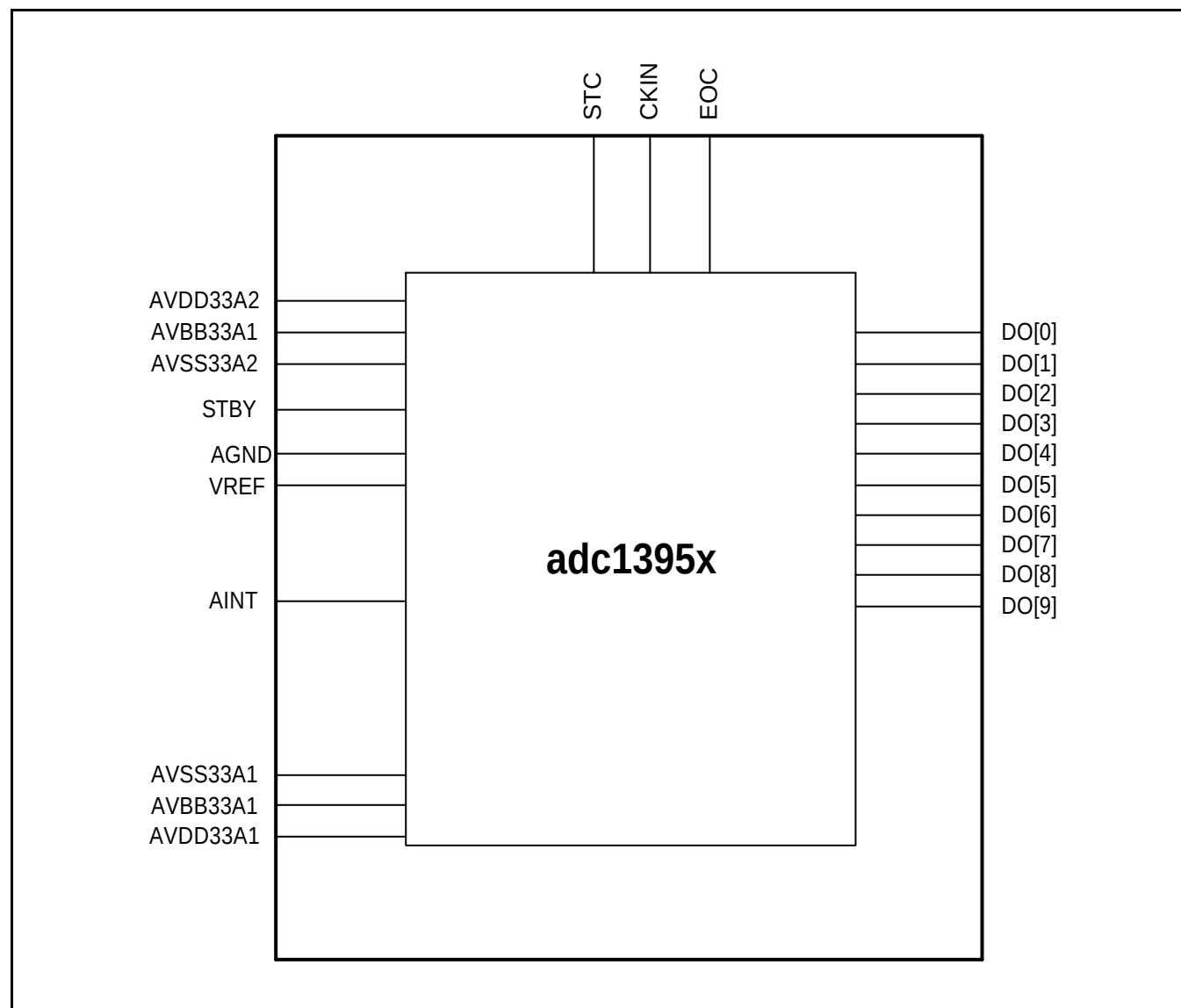
USER GUIDE

Input Range

The analog input is single-ended type and the range is from VREF to AGND. This AINT voltage follows reference voltage range fundamentally. So, if you want to alter into the another input range, you should change the voltage value of VREF.

You can use the AINT voltage whose minimum range is 2.0V. In this case, the VREF is 2.0V.

PHANTOM CELL INFORMATION



Name	I/O Type	Pin Usage	Pin Description
AINT	AI	Internal/External	AINT signal should not be crossed by any signals and should not run next to digital signals to minimize capacitive coupling between the two signals.
STBY	DI	Internal/External	Digital input signal lines must have same length to reduce propagation delay.
CKIN	DI	Internal/External	
DO[9:0]	DO	Internal/External	
EOC	DO	Internal/External	
STC	DI	Internal/External	
VREF	AI	External	Voltage reference lines (VREF and AGND) must be wide metal to reduce voltage drop of metal lines.
AGND	AI	External	
AVDD33A1	AP	External	It is recommended that you use thick analog power metal. When connected to PAD, the path should be kept as short as possible. Digital power and analog power are separately used.
AVBB33A1	AG	External	
AVSS33A1	AG	External	
AVSS33A2	DG	External	
AVDD33A2	DP	External	

HISTORY CARD

Version	Date	Modified Items	Comments
Ver1.0	02. 04	Original version published (preliminary)	
Ver1.1	04. 02	Addition of test results.	